GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Independent Set 5.02 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A figure with perimeter 18 cm is scaled by k = 3. Find the copy's perimeter.

54 cm

2. A figure with area 7 cm^2 is scaled by k = 3. Find the copy's area.

63 cm^2

Area × k^2 = 7 × 9.

3. A figure with area 40 cm^2 is scaled by k = ½. Find the copy's area.

10 cm^2

Area × $(½)^2$ = 40 × ¼.

4. A 2 cm by 5 cm rectangle is scaled by k = 2.5. Find the copy's perimeter and area.

P = 35 cm, A = 62.5 cm^2

P = 14 × 2.5; A = 10 × 6.25.

5. A figure has area 9 cm^2 and its scale copy has area 144 cm^2. Find k.

k = 4

144 ÷ 9 = 16 = k^2, so k = √16.

6. A figure's perimeter is 12 cm and its copy's perimeter is 48 cm. What is the area ratio, copy to original?

16

k = 4 from the perimeters; the area ratio is k^2 = 16.